26/03/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

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MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Test - I5

Time : 3.60 Hrs.

Answer Key

1.	(4)	37.	(1)	73.	(4)	109.	(1)	145.	(2)
2.	(2)	38.	(4)	74.	(2)	110.	(2)	146.	(1)
3.	(3)	39.	(4)	75.	(3)	111.	(3)	147.	(2)
4.	(3)	40.	(3)	76.	(1)	112.	(2)	148.	(4)
5.	(1)	41.	(1)	77.	(1)	113.	(2)	149.	(2)
6.	(3)	42.	(3)	78.	(1)	114.	(4)	150.	(2)
7.	(3)	43.	(1)	79.	(2)	115.	(4)	151.	(3)
8.	(2)	44.	(1)	80.	(4)	116.	(4)	152.	(4)
9.	(1)	45.	(2)	81.	(3)	117.	(4)	153.	(3)
10.	(1)	46.	(2)	82.	(3)	118.	(1)	154.	(1)
11.	(1)	47.	(3)	83.	(2)	119.	(1)	155.	(3)
12.	(2)	48.	(1)	84.	(1)	120.	(3)	156.	(2)
13.	(4)	49.	(4)	85.	(4)	121.	(1)	157.	(4)
14.	(4)	50.	(2)	86.	(3)	122.	(4)	158.	(4)
15.	(1)	51.	(1)	87.	(3)	123.	(2)	159.	(3)
16.	(4)	52.	(4)	88.	(4)	124.	(1)	160.	(4)
17.	(3)	53.	(2)	89.	(2)	125.	(2)	161.	(2)
18.	(2)	54.	(3)	90.	(1)	126.	(4)	162.	(4)
19.	(1)	55.	(3)	91.	(3)	127.	(1)	163.	(4)
20.	(2)	56.	(1)	92.	(1)	128.	(1)	164.	(2)
21.	(2)	57.	(4)	93.	(2)	129.	(4)	165.	(1)
22.	(2)	58.	(2)	94.	(4)	130.	(1)	166.	(2)
23.	(3)	59.	(2)	95.	(3)	131.	(3)	167.	(2)
24.	(3)	60.	(2)	96.	(1)	132.	(2)	168.	(3)
25.	(3)	61.	(2)	97.	(1)	133.	(2)	169.	(1)
26.	(2)	62.	(2)	98.	(2)	134.	(1)	170.	(1)
27.	(1)	63.	(1)	99.	(2)	135.	(4)	171.	(4)
28.	(3)	64.	(3)	100.	(1)	136.	(3)	172.	(1)
29.	(2)	65.	(4)	101.	(3)	137.	(3)	173.	(2)
30.	(4)	66.	(4)	102.	(4)	138.	(2)	174.	(1)
31.	(2)	67.	(1)	103.	(4)	139.	(1)	175.	(4)
32.	(1)	68.	(3)	104.	(2)	140.	(3)	176.	(1)
33.	(4)	69.	(2)	105.	(1)	141.	(3)	177.	(1)
34.	(3)	70.	(3)	106.	(1)	142.	(2)	178.	(3)
35.	(1)	71.	(1)	107.	(3)	143.	(3)	179.	(4)
36.	(1)	72.	(4)	108.	(3)	144.	(2)	180.	(4)

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Receptors in knee-jerk reflex are present in muscle spindle.

171. Answer (4)

Immuno-suppressant drugs are required even after tissue matching and blood group matching. Anti-histamine is given to patients suffering from

allergy. Colostrum provides passive natural immunity. Primary and secondary immune response is carried out by both B-lymphocytes and T-lymphocytes.

172. Answer (1)

Louis Pasteur by experimentation demonstrate that life comes only from pre-existing life and discarded the theory of spontaneous generation forever.

173. Answer (2)

Homologous organs :

- Heart of vertebrates
- Mouthparts of insects
- Forelimbs of whale, bats and cheetah
- Thorns of *bougainvillea* and tendrils of *cucurbita*

Analogous organs :

- Flippers of Penguins and Dolphins
- Eyes of Octopus and humans
- Potato and sweet potato.

174. Answer (1)

When average value is selected and extreme values are rejected, then such type of selection is called stabilizing selection.

175. Answer (4)

Ootid, spermatid and first polar body are haploid cells whereas primary oocyte is diploid cell.

176. Answer (1)

In unicellular organisms, cell division is itself a mode of reproduction.

177. Answer (1)

Sex is determined only by male gamete. Milk secretion starts towards end of pregnancy.

178. Answer (3)

Oestrogen acts on mammary glands for its growth and development. Lactation is under control of prolactin and progesterone.

179. Answer (4)

Trichomoniasis – Caused by protozoan.

AIDS and genital warts – Caused by viruses.

Syphilis – a bacterial disease caused by *Treponema pallidum*.

180. Answer (4)

Metal ions form coordination bonds for their activity with side chains at the active site and at the same time form one or more coordination bonds with the substrate e.g. zinc is a cofactor for proteolytic enzyme carboxypeptidase.

❑ ❑ ❑

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140. Answer (2)
Amount of air left in lungs after maximal forced expiration is called residual volume (1100-1200 mL).

150. Answer (2)
High pH, low H^+ ion concentration and low temperature are responsible for association of oxyhaemoglobin and shift the curve towards left.

151. Answer (3)
Haemoglobin consists of two parts, haem and globin. Haem consists of tetrapyrrole and iron.

152. Answer (4)
Opening of voltage gated Na^+ channels is responsible for depolarisation and opening of voltage gated K^+ channels is responsible for repolarisation phase in skeletal muscle fibres.

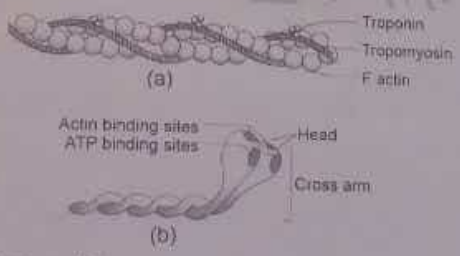
153. Answer (3)
B represents Q-wave of ECG.

154. Answer (1)
Greatest reabsorption of water from filtrate is in PCT, in the presence or absence of ADH.

155. Answer (3)
Kidneys are situated between levels of last thoracic and 3rd lumbar vertebra close to the dorsal inner wall of abdominal cavity.

156. Answer (2)
Ball and socket joint is present between humerus and glenoid cavity of scapula present in pectoral girdle.

157. Answer (4)



158. Answer (4)
Gaps between two adjacent myelin sheaths is called nodes of Ranvier found only in myelinated nerve fibres.

159. Answer (3)
Receptors in knee-jerk reflex are present in muscle spindle.

160. Answer (4)
The basal end of hair cells are in close contact with afferent nerve fibres and above the rows of hair cells is a thin elastic membrane called tectorial membrane.

161. Answer (2)
Addison's disease is caused due to hyposecretion of mineralocorticoids and glucocorticoids.

162. Answer (4)
Oestrogen is female sex hormone which has no effect on erythropoiesis.

163. Answer (4)
Hepatitis B is not curable even if diagnosed earlier and treated properly.

164. Answer (2)
Plasmid vector pBR322 contains two antibiotic resistance gene (tet^R and amp^R as selectable marker which are useful in selection of recombinant from non-recombinant and trans formant from non-transformant.

165. Answer (1)
Taq polymerase is obtained from thermophilic bacteria *Thermus aquaticus*. It is a thermostable DNA polymerase.

166. Answer (2)
MOET is meant for improving livestock.

167. Answer (2)
Graft rejection is mainly due to T-lymphocytes. Thymus gland provides micro-environment for the development and maturation of T-lymphocytes.

168. Answer (3)
MRI uses strong magnetic field and non-ionizing radiations to accurately detect pathological and physiological changes in the living tissue.

169. Answer (1)
Diacetylmorphine is obtained by acetylation of morphine and is commonly known as smack.

170. Answer (1)
Mucus coating of the epithelial living of GIT, urogenital tract and respiratory tract act as physical barrier which prevents the entry of micro-organisms. Epithelial living without mucus cannot act as a physical barrier.

171. Answer (4)
Immuno-suppressant drugs are required even after tissue matching and blood group matching. Anti-histamine is given to patients suffering from

allergy. Colostrum provides passive natural immunity. Primary and secondary immune response is carried out by both B-lymphocytes

175. Answer (4)
Ootid, spermatid and first polar body are haploid cells whereas primary oocyte is diploid cell.

127. Answer (1)

Organism is a basic unit of study in ecology.

128. Answer (1)

Natality and immigration contributes to an increase in population density.

129. Answer (4)

Deep sea is least productive ecosystem.

130. Answer (1)

Secondary succession starts on areas that somehow lost all the living organism that existed there.

It occurs in areas such as in abandoned farm lands, burned or cut forests, lands that have been flooded.

131. Answer (3)

There are 378 species of reptiles, and 3000 species of fishes found in Amazonian rain forest.

132. Answer (2)

Quagga is recently extincted from Africa.

133. Answer (2)

Colloidal materials present as impurities in waste water are faecal matter, bacteria, cloth and paper fibres.

134. Answer (1)

Joint forest management (JFM) was started by Government of India in 1980.

135. Answer (4)

Release of oxygen is associated with photochemical phase of photosynthesis.

ZOOLOGY

136. Answer (3)

Scypha is placed in phylum Porifera and represents cellular level of body organisation.

137. Answer (3)

Coelenterates are diploblastic acoelomate animals.

138. Answer (2)

Metameric segmentation is absent in molluscs.

139. Answer (1)

Presence of mammary glands is the most unique mammalian feature.

140. Answer (3)

Icthyophis is limbless amphibian. *Bufo* is commonly known as toad.

141. Answer (3)

A-Oesophagus – Passage for food

B-Salivary glands – Secretion of saliva

C-Hepatic caecae – Secrete digestive juice

D- Rectum – Opens out through anus

142. Answer (2)

On an average, females of cockroach produces 9-10 ootheca, each containing 14-16 fertilised eggs. Development takes place through number of nymphal stage having about 13 moults to reach adult stage.

143. Answer (3)

Hormones are secretions of endocrine glands. Milk is secretion of exocrine gland.

144. Answer (2)

In epithelial tissue, cells are compactly packed with little intercellular matrix.

145. Answer (2)

Lipase acts on lipids to form fatty acids and glycerol. Dipeptidase acts on dipeptides to form amino acids. Carboxypeptidase is an exopeptidase and acts on c-terminal of proteins and derived proteins to produce amino acids. Nucleases act on nucleic acids to produce nucleotides.

146. Answer (1)

Secretion of saliva is under control of both conditioned and unconditioned reflex actions. Secretion of gastric juice, pancreatic juice and intestinal juice is both under neural and hormonal controls.

147. Answer (2)

Increased liquidity of faecal discharge is known as diarrhoea.

148. Answer (4)

Gaseous exchange is the function of respiratory zone of the respiratory system (alveoli and their ducts).

105. Answer (1)
Ψ_w of cells, A, B and C are respectively –5, –6 and –3 bar.
Since, water moves from the point where water potential is greater, to the other point where water potential is less.
Hence, both cells A & B receives water from cell 'C' and water from cell 'A' moves towards cell 'B'.

106. Answer (1)
The process of absorption of water by hydrophilic solid particles of a substance without forming a solution is called imbibition.

107. Answer (3)
Bacteria (*Rhizobium*) initiates the nodule formation in the inner cortex region of root.

108. Answer (3)
Anabaena is a filamentous bacterium, which exhibits autotrophic mode of nutrition and fixes nitrogen symbiotically as well as in free-living state.

109. Answer (1)
In C_3 Pathway, the primary acceptor of CO_2 is RuBP; a 5 C – compound.

110. Answer (2)
Ps-II is found in grana lamellae only.

111. Answer (3)
Cytochrome c oxidase complex containing cytochrome a and a_3 and two copper centres is referred as complex IV.

112. Answer (2)
RQ of glucose, Tripalmitin, Malic acid and Oxalic acids are respectively 1.0, 0.7, 1.33 and 4.0.

113. Answer (2)
Gibberellins are composed of terpenes.

114. Answer (4)
Day neutral plants do not show only correlation between exposure to light duration and induction of flowering response. E.g., Tomato, Cucumber, Wheat and Radish are long day plants. Tabacco and Soyabean are short – day plants.

115. Answer (4)
Most of the species of *Chara* are monoecious in which the nucule occupies an upper position than globule.

116. Answer (4)
Zoospores produced in algae are asexual structure. These are usually naked but zygote is thin walled.

117. Answer (4)
In the majority of aquatic plants, flowers emerge above the level of water and are pollinated by insects or wind e.g., water hyacinth and water lily are pollinated by insects.

118. Answer (1)
Pollen grains are well-preserved as fossils because of the presence of sporopollenin (one of the most resistant organic materials known).

119. Answer (1)
A diploid organism when it is heterozygous for three loci and homozygous for two loci can have the genotype, AaBbCcDDEE.
Hence, type of gamete produced = 2^3
[$\because$ n = number of heterozygous loci]
$= 2^3 \Rightarrow 8$ [$\because n = 3$].

120. Answer (3)
In PKU, the affected individual lacks a liver enzyme phenylalanine hydroxylase that converts phenylalanine into tyrosine.

121. Answer (1)
AUG acts as initiator codon. It codes for methionine.

122. Answer (4)
Chromosome 1 has most genes (2968) while chromosome Y has the fewest genes (231) w.r.t human genome.

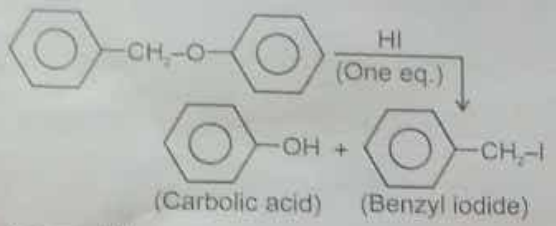
123. Answer (2)
Pusa snowball K-1 is a variety of cauliflower.

124. Answer (1)
Recovery of healthy plants from diseased plants is possible by meristem culture.

125. Answer (2)
Toddy left for few hours undergoes fermentation due to presence of naturally occurring yeast.

126. Answer (4)
A. Streptokinase → Clot buster.
B. Cyclosporin A → Immunosuppressive agent.
C. Statins → Blood-cholesterol lowering agent.
D. Lipase → Detergent formulation

86. Answer (3)



87. Answer (3)

2-ethylanthraquinol $\xrightarrow{O_2(air)}$ H_2O_2 + oxidised product

88. Answer (4)

In Clark's method, calculated amount of lime is added to remove temporary hardness of water.

89. Answer (2)

All enzymes that utilise ATP in phosphate transfer require Mg as the cofactor.

90. Answer (1)

Solubility order :

$MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

BOTANY

91. Answer (3)

The organelles included in endomembrane system are ER, Glogi complex, Lysosome and vacuoles.

Vacuole contains water, sap, excretory products and other material not useful for the cell.

92. Answer (1)

Ends of chromosomes are called telomeres. They seal the ends of chromosomes and prevent their shortening or chromosome loss.

93. Answer (2)

G_2 phase or Gap_2 phase is the second gap phase present between S and M phase of cell cycle. In this phase, tubulin protein synthesis takes place. Mitochondria, chloroplasts and Golgi bodies duplicates in G_2 phase.

94. Answer (4)

- Condensation and coiling of chromatin begins during Leptotene.
- During zygotene, activity of synaptonemal complex is seen.
- Homologous chromosome clearly appears as tetrad during pachytene.
- Terminalisation of chiasmata is seen in diakinesis.

95. Answer (3)

Brinjal belongs to class – Dicotyledonae.

96. Answer (1)

Key: Identification of plants and animals based on similarities and dissimilarities.

Monograph: Contains information of any one taxon.

Botanical garden: *Ex-situ* conservation strategies of plants.

Museum: Collection of preserved plants and animals.

97. Answer (1)

In members of oomycetes, hyphal wall contains cellulose and other glucans. *e.g.*, *Albugo*, *Phytophthora*, *Pythium*, etc.

98. Answer (2)

Members of kingdom Monera and Protista exhibits both autotrophic as well as hetetrotrophic mode of nutrition.

99. Answer (2)

The arrangement of flowers on the floral axis of the plant is known as 'Inflorescence'.

100. Answer (1)

The hilum is a scar on the seed coat through which the developing seeds were attached to fruit.

101. Answer (3)

In dicot leaf, vascular bundles differ in size due to presence of reticulate venation.

102. Answer (4)

Conjoint closed vascular bundles is found in monocot stem.

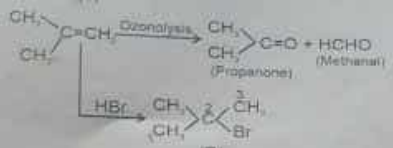
103. Answer (4)

Laminaria and *Fucus* are rich source of iodine.

104. Answer (2)

Members of gymnosperm have archegonia as well as they produce seeds, *e.g.*, *Pinus*, *cedrus*, *Ginkgo*.

70. Answer (3)



2-bromo-2-methylpropane

71. Answer (1)

$$X_{H_2O} = \frac{n_{H_2O}}{n_{NaOH} + n_{H_2O}} = \frac{\frac{90}{18}}{\frac{10}{40} + \frac{90}{18}} = 0.95$$

72. Answer (4)

$$\% \text{ of Mg} = \frac{\text{Mass of Mg}}{\text{Mass of compound}} \times 100$$

$$0.2 = \frac{24 \times 100}{MW_{minimum}}$$

$\Rightarrow MW_{minimum} = 1.2 \times 10^4$

73. Answer (4)

Orbital angular momentum = $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

and $\ell = 1$ for p-subshell.

74. Answer (2)

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.626 \times 10^{-34}}{10^{-6} \times 1000} = 6.626 \times 10^{-31} m$$

75. Answer (3)

Tolerable limit for NO_3^- and SO_4^{2-} are 50 ppm and 500 ppm respectively.

76. Answer (1)

Metallic character : Si < Mg < Na

77. Answer (1)

Due to large size of Cl, HCl does not show H-bonding.

78. Answer (1)

I_3^- : sp^3 hybridised $\Rightarrow$ Bent species

79. Answer (2)

According to MOT, C_2 contains two π-bonds.

80. Answer (4)

$$A_2B_2(s) \rightleftharpoons 2AB(g) + B(g)$$

2P atm P atm

$P_{Total} = 2P + P = 3P = 3$ atm

$P = 1$ atm

$$K_p = \frac{(p_{AB})^2 (p_B)}{1} = (2)^2 (1) = 4 \text{ atm}^3$$

81. Answer (3)

A : $8 \times \frac{1}{8} = 1$, B : $1 \times 1 = 1$,

C : $6 \times \frac{1}{2} + 12 \times \frac{1}{4} = 6$

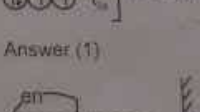
$\therefore$ Formula = ABC_6

82. Answer (3)

- Higher the value of (i × M), more will be the boiling point of aqueous solution.
- Solute: Urea (i = 1), NaCl (2), Na_2SO_4 (3), NH_4NO_3 (2)

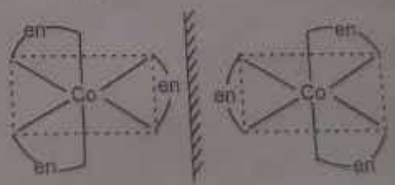
83. Answer (2)

E.C. of Co^{3+} in $[CoF_6]^{3-}$ will be



As F are weak field ligands

84. Answer (1)



(Non-superimposable mirror images)

85. Answer (4)

More the stability of carbocation intermediate,

86. Answer (3)



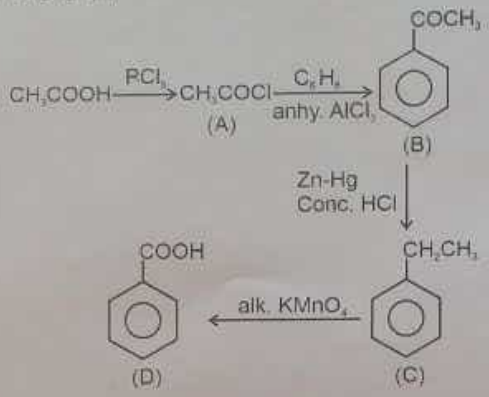
88. Answer (4)

In Clark's method, calculated amount of lime is added to remove temporary hardness of water.

Arginine is an essential amino acid.

ketones containing at least two α-H atoms.

56. Answer (1)

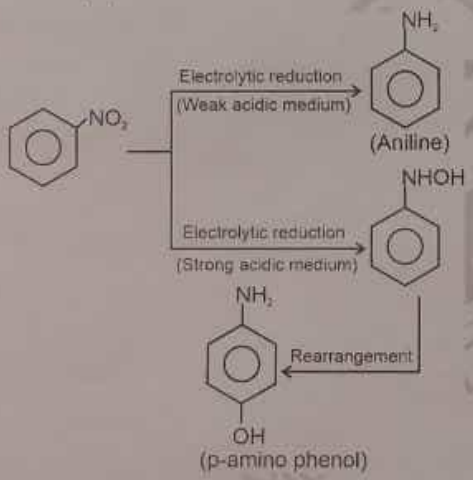


57. Answer (4)

Due to combination of steric and hydration effect, the basic strength order of ethyl amines is

$2^\circ > 3^\circ > 1^\circ$

58. Answer (2)



59. Answer (2)

Malachite : $CuCO_3.Cu(OH)_2$

Siderite : $FeCO_3$

Cryolite : Na_3AlF_6

Sphalerite : ZnS

60. Answer (2)

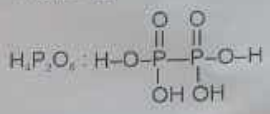
Bond energy : $Cl_2 > Br_2 > F_2$

Due to lp-lp repulsion, bond energy of F_2 is less.

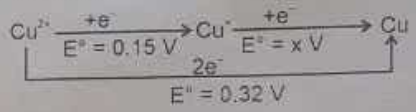
61. Answer (2)

MnO_4^- contains one unpaired electron in d-orbital of Mn.

62. Answer (2)



63. Answer (1)



$\therefore \Delta G_3 = \Delta G_1 + \Delta G_2$

$-2 \times 0.32 \times F = -1 \times 0.15 \times F - 1 \times x \times F$

$x = +0.49$ V

64. Answer (3)

$\Lambda_m^\circ[Ba(OH)_2] = \Lambda_m^\circ(BaCl_2) - 2\Lambda_m^\circ(NaCl) + 2\Lambda_m^\circ(NaOH)$

$= z - 2x + 2y$

65. Answer (4)

Rate of reaction

$$-\frac{d[A]}{dt} = -\frac{1}{2}\frac{d[B]}{dt} = \frac{1}{4}\frac{d[C]}{dt} = \frac{1}{3}\frac{d[D]}{dt}$$

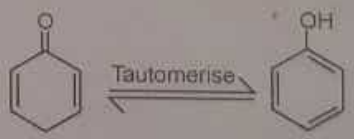
66. Answer (4)

Sb_2S_3 sol is negatively charged.

67. Answer (1)


contains sp^3 hybridised carbon.

68. Answer (3)



- Bridge H-atom does not take part in tautomerism.

69. Answer (2)

Activating nature :

$-OH > -NHCOCH_3 > -CH_3 > -Cl$

$$\phi = \phi_0 + \frac{1}{2}\left(\frac{2\phi - \phi_0}{2}\right)$$

$$\phi = \phi_0 + \frac{\phi}{2} - \frac{\phi_0}{4}$$

$$\frac{\phi}{2} = \frac{3\phi_0}{4}$$

$$\phi_0 = \frac{2\phi}{3}$$

$$\phi_0 = \frac{2h\nu}{3}$$

40. Answer (3)

$$L = \frac{nh}{2\pi}$$

41. Answer (1)

42. Answer (3)

By conservation of energy

$$\frac{1}{2}mv^2 = \frac{2e \times 2e}{4\pi\varepsilon_0 r}$$

$$r = \frac{2e^2}{\pi\varepsilon_0 m}$$

$$r \propto \frac{1}{m}$$

43. Answer (1)

When an impurity is dopped into an intrinsic semiconductor, the conductivity of semiconductor increases

44. Answer (1)

Both diodes will be in forward biased

$$I = \frac{V}{R_{eff}}$$

$$= \frac{10}{2} = 5\ A$$

45. Answer (2)

$Y = A\ \bar{A} = 0$

For any value of A, Y will be zero.

CHEMISTRY

46. Answer (2)

Gas	H_2	He	O_2	N_2
$T_C(K)$	33.2	5.3	154.3	126

47. Answer (3)

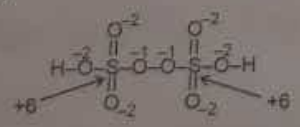
When $(n_g)_P > (n_g)_R$, entropy increases.

48. Answer (1)

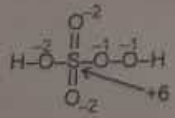
A weak acid with its conjugate base behaves as an acidic buffer.

49. Answer (4)

$H_2S_2O_8$:



H_2SO_5 :



50. Answer (2)

Arginine is an essential amino acid.

51. Answer (1)

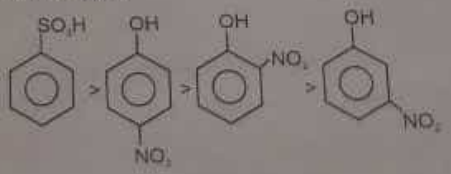
Neoprene is polymer of 2-chlorobuta-1,3-diene

52. Answer (4)

Alitame is 2000 times sweeter than cane sugar.

53. Answer (2)

Acidic nature



54. Answer (3)

Reactivity order towards nucleophilic addition reaction.

HCHO > CH_3CHO >  > CH_3COCH_3

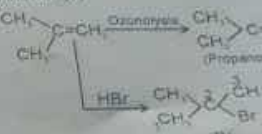
55. Answer (3)

Aldol condensation is given by aldehydes and ketones containing at least two α-H atoms.

56. Answer (1)

61. Answer (2)

70. Answer (3)



2-bromo-2-methy

71. Answer (1)

$$X_{H_2O} = \frac{n_{H_2O}}{n_{NaOH} + n_{H_2O}} = \frac{9}{10}$$

72. Answer (4)

$$\%\ \text{of Mg} = \frac{\text{Mass of M}}{\text{Mass of com}}$$

$$0.2 = \frac{24 \times 100}{MW_{minimum}}$$

$\Rightarrow MW_{minimum} = 1.2 \times$

73. Answer (4)

Orbital angular mom

and $l = 1$ for p-subs

74. Answer (2)

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.626 \times 10^{-34}}{10^{-5} \times 1000}$$

75. Answer (3)

Tolerable limit

and 500 ppm

76. Answer (1)

Metallic chara

77. Answer (1)

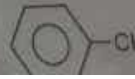
Due to larg

H-bonding.

78. Answer (1)

I_3^- : sp^3 hyb

86. Answer (3)



$\phi = BA$

$= \frac{2\sqrt{2}\mu_0 I}{4\pi L} I^2$

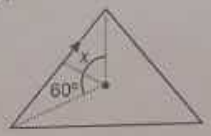
$\phi = MI$

$M \propto \frac{I^2}{L}$

30. Answer (4)

Fact based

31. Answer (2)



$B = 3\left(\frac{\mu_0 I}{4\pi x}\right)(\sin 60° + \sin 60°)$

$B = 3\left(\frac{\mu_0 I}{4\pi\left(\frac{a}{2\sqrt{3}}\right)}\sqrt{3}\right)$

$= \frac{9\mu_0 I}{2\pi a}$

32. Answer (1)

$\omega_0 = \frac{1}{\sqrt{LC}}$

$X_L = 2\omega_0 L$

$= 2\sqrt{\frac{L}{C}}$

$X_C = \frac{1}{2}\frac{1}{\omega_0 C}$

$= \frac{1}{2\sqrt{\frac{C}{L}}}$

$= \frac{1}{2}\sqrt{\frac{L}{C}}$

$\frac{X_L}{X_C} = 4 : 1$

33. Answer (4)

Maxwell's equation describes the fundamental law of both electricity and magnetism.

34. Answer (3)

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\lambda_r > \lambda_v \Rightarrow \mu_r < \mu_v \Rightarrow f_r > f_v$

35. Answer (1)

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$M = -\frac{v}{u}$

$\frac{1}{2} = \frac{v}{u}$

$v = \left(\frac{u}{2}\right)$

$\frac{1}{\frac{u}{2}} - \frac{1}{u} = \frac{1}{2}$

$\frac{2}{u} - \frac{1}{u} = \frac{1}{2}$

$u = 2m$

36. Answer (1)

$W' = \frac{W}{\mu}$

37. Answer (1)

$a\sin\theta = n\lambda$

$\Rightarrow \frac{ay}{D} = \lambda \Rightarrow a = \frac{D\lambda}{y}$

$a = \frac{2 \times 6000 \times 10^{-10}}{4 \times 10^{-3}}$

$a = 0.3$ mm

38. Answer (4)

$\lambda = \frac{h}{\sqrt{2mk}}$

$k \rightarrow 2$ times

$\lambda' = \frac{\lambda}{\sqrt{2}}$

39. Answer (4)

$\phi = \phi_0 + \frac{1}{2}mv^2$

$2\phi = \phi_0 + 2mv^2$

Code-A

18. Answer (2)
Coefficient of volume expansion of water will be negative between 0°C to 4°C

19. Answer (1)

$P_1 = \frac{nRT}{V}$

$P_2 = \frac{nRT}{V}$

$P_{net} = 2n\frac{RT}{V} = 2P$

20. Answer (2)

$P = aA^4$

21. Answer (2)
In adiabatic process

$Q = 0$

$W = -\Delta U = -nC_v\Delta T$

if $W > 0 \Rightarrow \Delta T < 0$

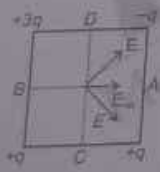
22. Answer (2)
Both spheres are identical therefore at equal potential charge on the each sphere will be

$q = \frac{q_1 + q_2}{2}$

$= \frac{-60e + 20e}{2}$

$= -20e$

23. Answer (3)



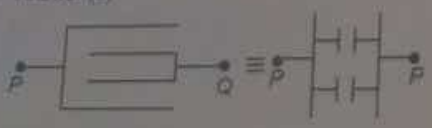
24. Answer (3)

$W = U_f - U_i$

$= -PE\cos180 + PE\cos0$

$= 2PE$

25. Answer (3)



$C_{eq} = 2C$

$= \frac{2A\varepsilon_0}{d}$

26. Answer (2)
Fractional increase in resistivity per unit increase in temperature is called temperature coefficient of resistivity

$\alpha = \frac{\Delta R}{R\Delta t}$

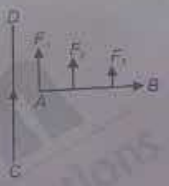
27. Answer (1)

$U = \frac{1}{2}CV^2$

$= \frac{1}{2}\times 2\times(2)^2$

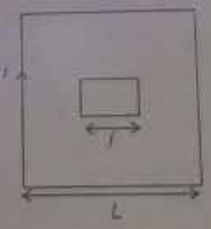
$= 4\ \mu J$

28. Answer (3)



Since magnetic field due to current in various location of AB is not uniform, therefore, the wire AB carrying current I_1 is subjected to variable magnetic field due to which neither torque nor force on the wire will be zero as a result of which wire AB will have both translational & rotational motion.

29. Answer (2)



$B = 4\left[\frac{\mu_0 I}{4\pi\left(\frac{L}{2}\right)}(\sin45 + \sin45)\right]$

$B = \left[\frac{\mu_0 I2\sqrt{2}}{\pi L}\right]$

$\phi = \phi_0 + \frac{1}{2}\left(\frac{2\phi - \phi_0}{2}\right)$

$\phi = \phi_0 + \frac{\phi}{2} - \frac{\phi_0}{4}$

$\frac{\phi}{2} = \frac{3\phi_0}{4}$

$\phi_0 = \frac{2\phi}{3}$

$\phi_0 = \frac{2hv}{3}$

40. Answer (3)

$L = \frac{nh}{2\pi}$

41. Answer (1)

42. Answer (3)
By conservation

$\frac{1}{2}mv^2 = \frac{2e\times 2e}{4\pi\varepsilon_0 r}$

46. Answer (2)

Gas	H_2
T_C(K)	33.2

47. Answer (3)
When $(n_0)e >$

48. Answer (1)
A weak acid
an acidic bu

49. Answer (4)

$H_2S_2O_8$:

H_2SO_4 :

50. Answer
Arginin

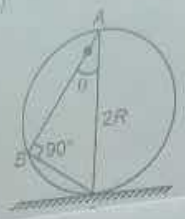
$\phi = BA$

$= \frac{2\sqrt{2}\mu_0 I}{4\pi L}I^2$

34. Answer (3)

$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

4. Answer (3)



$\frac{1}{2}mv^2 = 2$

$v = \sqrt{7gr}$

7. Answer (3)

The coefficient of restitution depends upon the material of the body

8. Answer (2)

$|\vec{A}\times\vec{B}|^2 + |\vec{A}\cdot\vec{B}|^2 = (AB\sin\theta)^2 + (AB\cos\theta)^2$

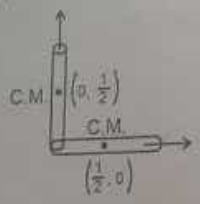
$= (AB)^2(\sin^2\theta + \cos^2\theta)$

$= (AB)^2$

$= (2\times 3)^2$

= 36 units

9. Answer (1)



$X_{C.M} = \frac{\frac{1}{4}\times\frac{m}{2} + \frac{0.m}{2}}{m}$

$X_{C.M} = \frac{1}{8}$

$Y_{C.M} = \frac{0\times\frac{m}{2} + \frac{m}{2}\times\frac{1}{4}}{m} = \frac{1}{8}m$

$\vec{r} = \frac{1}{8}\hat{i} + \frac{1}{8}\hat{j}$

$|\vec{r}| = \frac{\sqrt{2}}{8} = \frac{1}{4\sqrt{2}}m$

10. Answer (1)

$\frac{\text{Translational K.E}}{\text{Rotational K.E}} = \frac{\frac{1}{2}mV^2}{\frac{1}{2}I\frac{V^2}{R^2}}$

$\frac{3}{2} = \frac{m}{\frac{I}{R^2}}$

$I = \frac{2mR^2}{3}$ (Hollow sphere)

11. Answer (1)

$V_{max} = A\omega$

$a_{max} = -\omega^2 A$

12. Answer (2)

$\Delta\phi = \omega_0\Delta t$

$= \frac{2\pi}{4}\times\frac{3}{4} = \frac{3\pi}{8}$ (rad)

13. Answer (4)

$V = \sqrt{\frac{GM}{r}}$

$V_0 = \sqrt{\frac{GM}{3R}}$

$V_0 = \sqrt{\frac{gR}{3}}$

14. Answer (4)

$\frac{\lambda_m}{2} = l \Rightarrow \lambda_{max} = 120$ cm (vibrating in one loop)

15. Answer (1)

$f_1 = \frac{3v}{4l}$

$60 = \frac{3v}{4}$

$\frac{v}{l} = 80$

$\frac{7v}{4l} = \frac{7}{4}\times 80 = 140$ Hz

16. Answer (4)

$W = 2\pi rT$

17. Answer (3)

$M = \rho V$

$0 = \Delta\rho V + \rho\Delta V$

$\frac{\Delta V}{V} = \frac{\Delta\rho}{\rho}$

$K = \frac{-P}{\frac{\Delta V}{V}}$

$-\frac{\Delta V}{V} = \frac{P}{K}$

$\Delta\rho = \frac{P\rho}{K}$

Aakash Medical IIT-JEE Foundations

Code-A

28/03/2021

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3:00 Hrs.

Test - 15

Hints and Solutions

PHYSICS

1. Answer (4)
The contribution of S. Chandrasekhar to physics is structure and evolution of star

2. Answer (2)
$X \sim L T^{-2}$
$\frac{\Delta X}{X} \times 100 = \frac{\Delta L}{L} \times 100 + 2\frac{\Delta T}{T} \times 100$
$= 0.2 + 2 \times 3$
$= 6.2\%$

3. Answer (3)
Suppose t is the time of fall
$h = \frac{1}{2}gt^2$ —(i)
$\frac{h}{2} = \frac{1}{2}g(t-1)^2$ —(ii)
On solving (i) & (ii)
$t = (2+\sqrt{2})s$

4. Answer (3)



$S = \frac{1}{2}at^2$
$AB = \frac{1}{2}(g\cos\theta)t^2$
$2R\cos\theta = \frac{1}{2}g\cos\theta t^2$
$t = 2\sqrt{\frac{R}{g}}$

5. Answer (1)
$\mu = 0.5 \quad N = 2mg(\sin\theta - \mu\cos\theta)$
$mg\cos\theta = 2mg(\sin\theta - \mu\cos\theta)$
$\cos\theta = 2(\sin\theta - \mu\cos\theta)$
$\frac{1}{2} = \tan\theta - \frac{1}{2}$
$\tan\theta = 1 \Rightarrow \theta = 45°$

6. Answer (3)
$\frac{1}{2}mv^2 = 2mrg + \frac{1}{2}m\left(\sqrt{3gr}\right)^2$
$\frac{1}{2}mv^2 = \frac{7mgr}{2}$
$v = \sqrt{7gr}$

7. Answer (3)
The coefficient of restitution depends upon the material of the body

18. Answer (2)
Coefficient of volume expansion of
negative between 0°C to 4°C

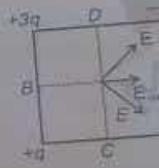
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$P_1 = \frac{nRT}{V}$
$P_2 = \frac{nRT}{V}$
$P_{net} = 2n\frac{RT}{V} = 2P$

20. Answer (2)
$P = \sigma AT^4$

21. Answer (2)
In adiabatic process
$Q = 0$
$W = -\Delta U = -nC_v\Delta T$
If $W > 0 \Rightarrow \Delta T < 0$

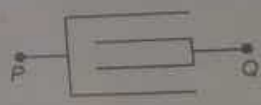
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Both spheres are identical th
potential charge on the each sp
$q = \frac{q_1 + q_2}{2}$
$= \frac{-60e + 20e}{2}$
$= -20e$

23. Answer (3)



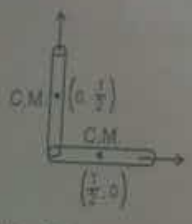
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$= -PE\cos180 + PE\cos0$
$= 2PE$

25. Answer (3)



8. Answer (2)
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$= (AB)^2(\sin^2\theta + \cos^2\theta)$
$= (AB)^2$
$= (2\times3)^2$
$= 36$ units

9. Answer (1)



1 m 0.m

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$\Delta\phi = \omega\Delta t$
$= \frac{2\pi}{4}\times\frac{3}{4} = \frac{3\pi}{8}$ (rad)

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$V_0 = \sqrt{\frac{GM}{3R}}$
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$\phi = BA$
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$\phi = Mi$
$M \propto \frac{l^2}{L}$

30. Answer (4)
Fact based

31. Answer (2)